

Houston Fire Department Data report



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Presented by

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Topics that will be discussed

1 Emt & Fire Events by Station



2 EMS vs Fire events Comparison



3 Top 10 Busiest stations by EMS



4 Top 10 Stations by Fire Events



5 Top 10 slowest stations (EMS)



6 Top 10 slowest by Fire events



Created by Isaac Hocker

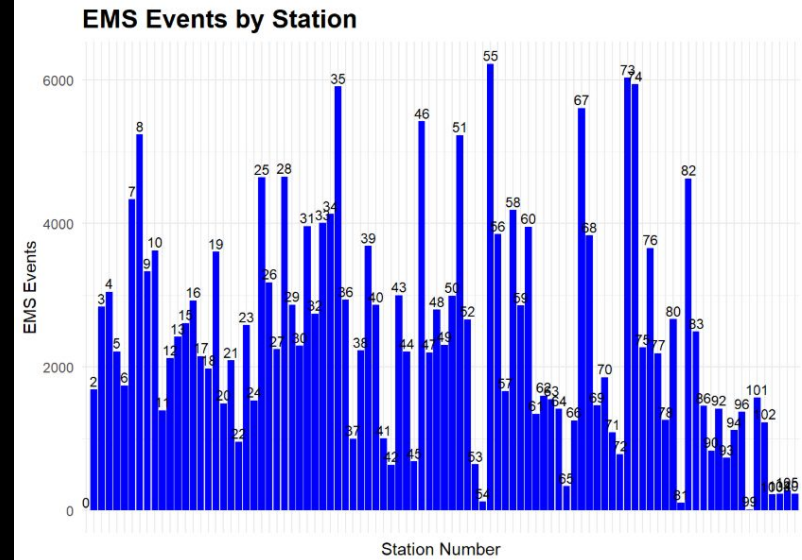
For my project I wanted to use data from Houston Fire Department from the year 2023 which would showcase incident reports based on EMS and Fire rescue calls. The reason I chose this project was because around this time last year I was previously enrolled in the Houston Fire Academy where I wanted to become a firefighter. The inspiration of becoming one was from my step father who became a firefighter about 11 years ago. Due to some circumstances and challenges I faced I was not able to get passed the EMT portion of the school which was the hardest. Basically a 6 month program stuffed into 2 and a half months of school. Ever since then I've had an appreciation for first responders not just fire fighters, but also police officers as well. Even though I am not covering hpd hopefully my project gives you an insight about what all first responders go through.

The Difference between EMS & Fire rescue

EMS calls are usually higher volume than fire rescue call due to the severity of fire events that happen. Usually EMS calls can range from someone having trouble breathing, to a stroke and even a heart attack. They also respond to calls where there were car accidents and fata shootings. While Fire rescue usually go to calls where there are burning buildings , but also calls where there might be a gas leak and a smoke detector going off.

Emt Events

This chart shows how many EMS events each station handled during the year of 2023. The taller bars represent the stations who had higher number of call volumes compared to the other stations with the lower numbers. The reason this data is important is because it highlights which areas based of the stations have highest emergency demand.

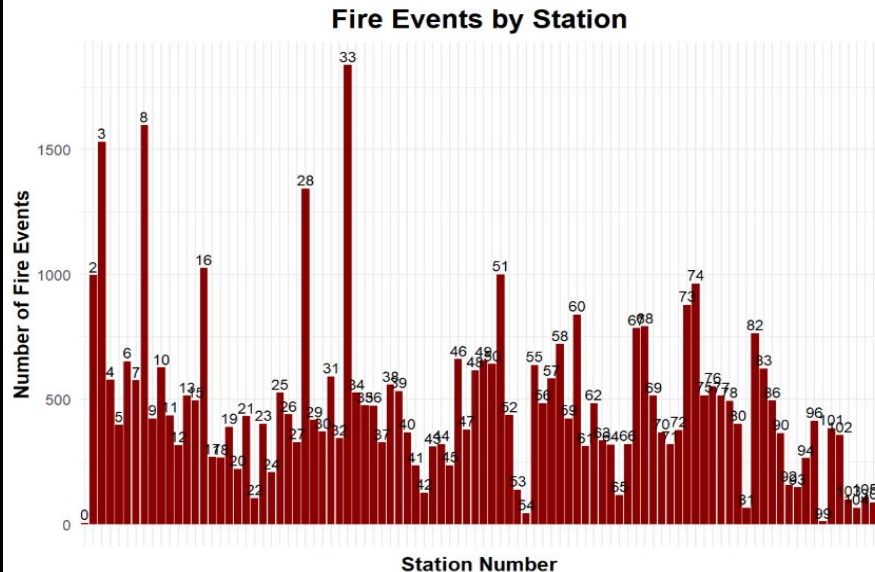


Emt events

```
hfd %>%
  filter(!is.na(EMS.Events), EMS.Events > 0) %>% # remove stations with no incidents
  mutate(Station = factor(Station)) %>% # make stations discrete that do not have activity
  ggplot(aes(x = Station, y = EMS.Events)) +
  geom_col(fill = "blue") +
  geom_text(aes(label = Station), vjust = -0.3, size = 3, color = "black") +
  labs(title = "EMS Events by Station", x = "Station Number", y = "EMS Events") +
  theme_minimal() +
  theme(axis.text.x = element_blank(), # remove number on the x axis
        axis.ticks.x = element_blank(), # remove tick marks
        plot.title = element_text(size = 16, face = "bold"))
)
```

Fire Events

The graph helps identify which stations that deal with the highest amount of fire activity. Knowing which stations are busier can help as a being a civilian in the outside we can determine which areas have more emergencies which can cause heavy traffic. Most of the time fires can start from numerous factors such as population density, older buildings with old wiring , and neighborhoods with higher risk factors.

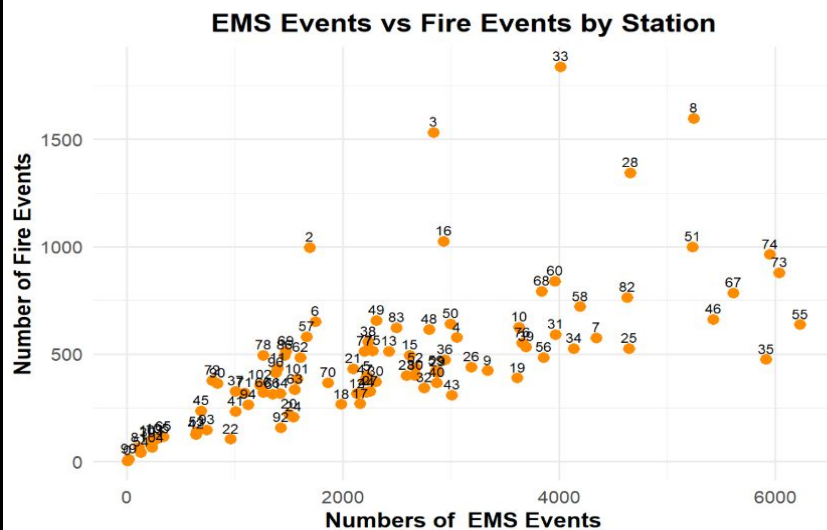


Fire events by station

```
hfd %>%
  filter(Fire.Events > 0) %>% # Here i removed stations with zero events
  mutate(Station = factor(Station)) %>%
  ggplot(aes(x = Station, y = Fire.Events)) +
  geom_col(fill = "darkred") +
  geom_text(aes(label = Station), vjust = -0.3, size = 3, color = "black") +
  labs(title = "Fire Events by Station", x = "Station Number", y = "Number of Fire Events") +
  theme_minimal() +
  theme(axis.text.x = element_blank(), # remove x axis numbers
        axis.ticks.x = element_blank(), # remove ticks
        plot.title = element_text(size = 16, face = "bold", hjust = 0.5),
        axis.title = element_text(size = 12, face = "bold"))
  )
```

Ems Events vs Fire events

This graph shows how the relationship between two types of emergency activities. Seeing how these events move together helps identify stations that face heavier workloads and may require additional support such as staffing. In a real emergency operations, strong relationship between both EMS and fire can help not just each other but us as well. Such as resolving incidents faster and especially safer for everyone else.

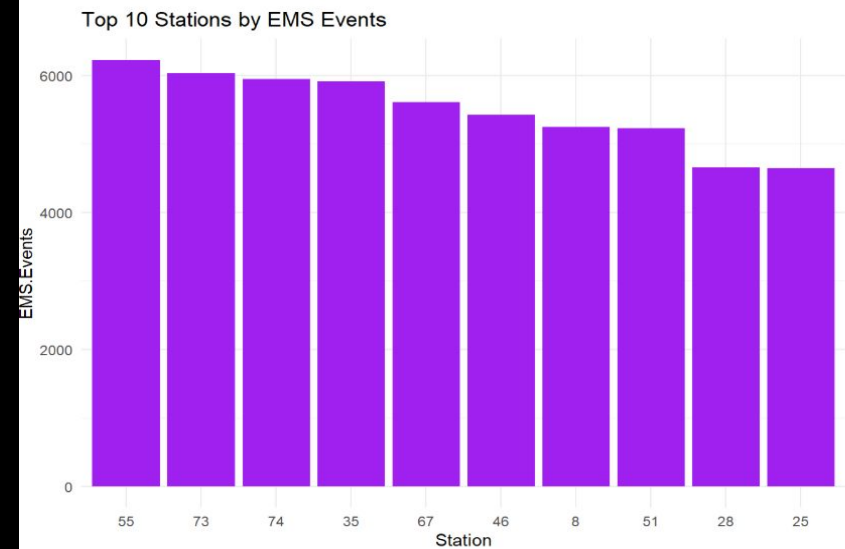


EMS Events vs Fire Events

```
hfd %>%
  filter(EMS.Events > 0, Fire.Events > 0) %>% # I dropped stations that had no activity.
  ggplot(aes(x = EMS.Events, y = Fire.Events, label = Station)) +
  geom_point(color = "darkorange", size = 3) + # slightly smaller points
  geom_text(vjust = -0.6, size = 3, color = "black") +
  labs(title = "EMS Events vs Fire Events by Station", x = "Numbers of EMS Events",
        y = "Number of Fire Events") +
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5),
        axis.title = element_text(face = "bold"))
)
```

Top 10 busiest stations by EMS calls

The graph on the right shows 10 of the busiest stations by EMS call volumes. The reason this is so important is so we can see which areas are the busiest and also seeing based on your eyes where we can add additional staffing or equipment based on how busy the stations are.

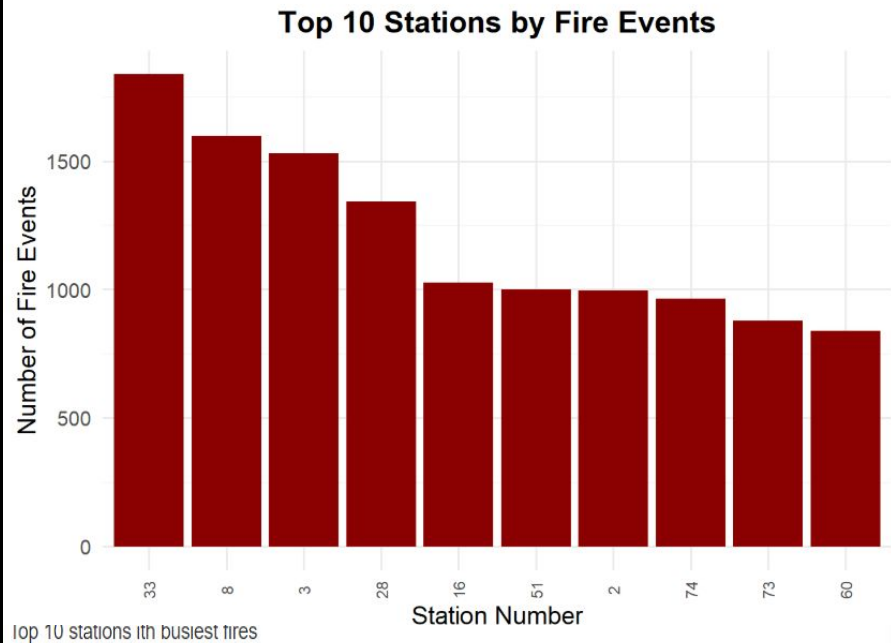


EMS Top 10 Busiest Stations for EMS

```
hfd %>%  
  arrange(desc(EMS.Events)) %>%  
  slice(1:10) %>%  
  mutate(Station = factor(Station, levels = Station)) %>%  
  ggplot(aes(x = Station, y = EMS.Events)) +  
  geom_col(fill = "purple") +  
  labs(title = "Top 10 Stations by EMS Events") +  
  theme_minimal()
```

Busiest stations based on fire call volumes

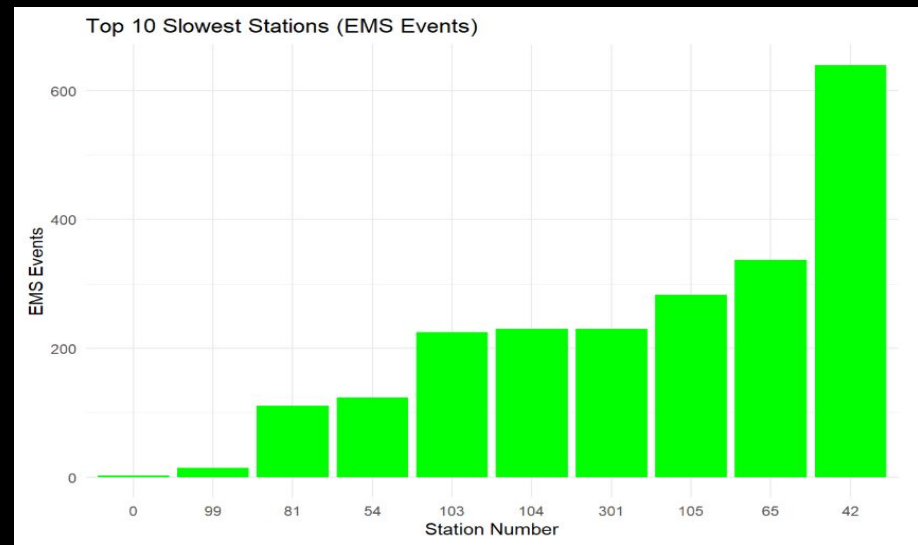
In a real emergency operations, stations with heavy fire call volumes face more stress on personnel, equipment, and response times. Identifying these stations can help the department also better manage workloads and prepare for peak times when these calls do happen. High fire activity can indicate areas where prevention programs or building inspections could be placed to reduce risk.



```
hfd %>%
  arrange(desc(Fire.Events)) %>% # sorting stations
  slice(1:10) %>% # keeping 1 through 10
  mutate(Station = factor(Station, levels = Station)) %>% # keep the bars in order
  ggplot(aes(x = Station, y = Fire.Events)) +
  geom_col(fill = "darkred") +
  labs(title = "Top 10 Stations by Fire Events", x = "Station Number", y = "Number of Fire Events") +
  theme_minimal(base_size = 14) +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, size = 9),
        plot.title = element_text(face = "bold", hjust = 0.5))
```

Slowest stations based on EMS calls

For this data I wanted to see where are the slowest stations based off EMS calls would be. The graph on the right helps determine if we need to put less resources in these stations and focus more on the busier stations. This could also benefit us as civilians to see which areas have lower risk of car accidents, and shootings that may happen as well. It's safe to assume that where there are lower call volumes are generally safer areas.

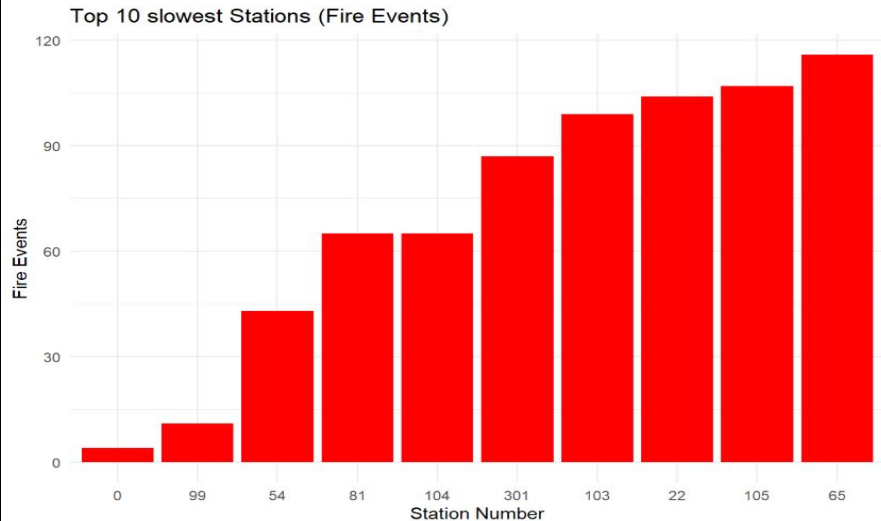


Slowest stations based on EMS calls

```
hfd %>%  
  arrange(EMS.Events) %>%  
  slice(1:10) %>%  
  mutate(Station = factor(Station, levels = Station)) %>%  
  ggplot(aes(x = Station, y = EMS.Events)) +  
  geom_col(fill = "blue") +  
  labs(title = "Top 10 Slowest Stations (EMS Events)", x = "Station Number", y = "EMS Events") +  
  theme_minimal()
```

Slowest stations based on Fire rescue calls

As I did with EMS I wanted to do Fire calls as well to show if any of the stations share the same volume based on EMS calls and from the graphs of both we can see which ones don't need as much staffing compared those with the highest call volumes.



Slowest stations based on fire calls

```
hfd %>%  
  arrange(Fire.Events) %>%  
  slice(1:10) %>%  
  mutate(Station = factor(Station, levels = Station)) %>%  
  ggplot(aes(x = Station, y = Fire.Events)) +  
  geom_col(fill = "red") +  
  labs(title = "Top 10 slowest Stations (Fire Events)", x = "Station Number", y = "Fire Events") +  
  theme_minimal()
```

Limitations

The data set only reflects only time period and not long term trends, basically because it is based off 2023 year data doesn't show the years after if there was and differences.

The data is aggregated(formed) by stations and events that happened. It does not include detailed call level events.

The data also does not specify what type of calls were specifically made but based on two categories which were Fire events and EMS events.



What we can learn from this data

This data gives an insight on how busy or how slow some stations are, which opens the eyes to those who might not be aware of how busy our first responders can be from a daily basis.

We can learn and benefit from this as well because the data shows each fire station and each fire station is located in a district that they monitor. How we can benefit from this is predetermining where heavy traffic or accidents occur and trying to avoid them. Also if you plan on moving to another area you can see which areas are typically safer compared to others.



In conclusion

This analysis of Houston Fire Department incident data highlights a big difference in EMS and Fire emergencies across all stations. Several stations consistently reported higher call volumes, suggesting concentrated emergency demand in those areas. Understanding these pattern is useful in operational plannings, risk assessment, and where to put resources. While the data does not include detailed call types or precise geographical information, it still provides valuable insight into overall workload distribution.

